

INTRODUCTION TO AGRICULTURE

The term agriculture is derived from the **Latin word *ager/agri* meaning 'soil' & *cultura* meaning 'cultivation'.**

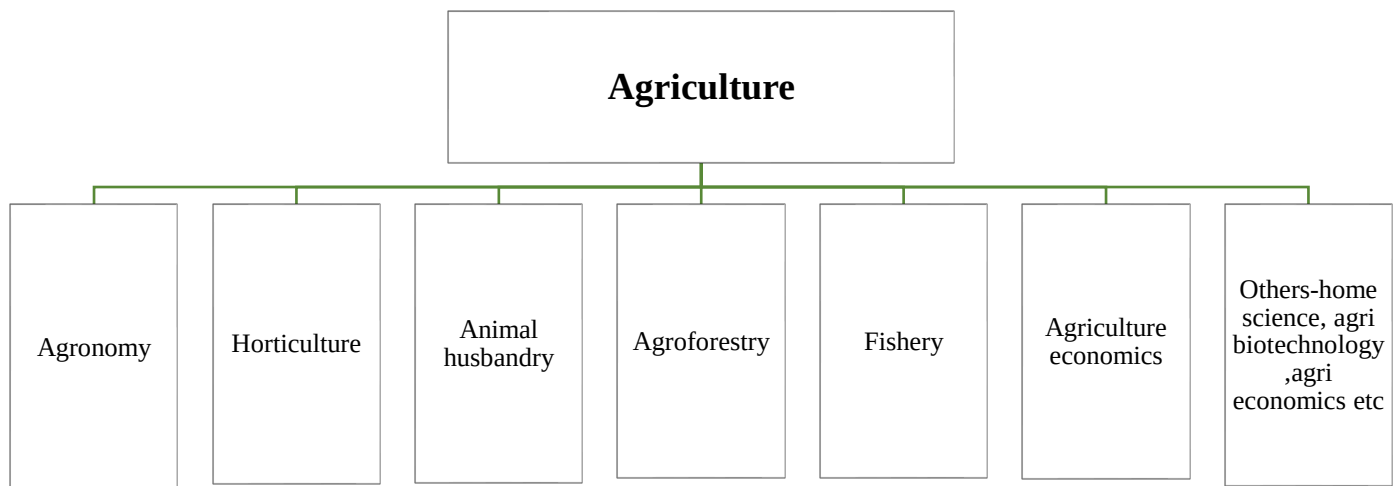
Definition

Agriculture is a very broad term **encompassing all aspects of crop production**, livestock farming, fisheries, horticulture etc.

Origin of agriculture

There **is no single or simple origin of agriculture** but the earliest evidence of planned sowing and harvesting of crops is in Egypt & India.

Branches of agriculture



AGRONOMY

This term is derived from the **Greek word *agros* meaning 'field' & *nomos* meaning 'to manage'.** **Agronomy is a branch of agriculture which deals with the principles & practices of soil, water, and crop management & to provide the favorable environment to the crop for getting higher productivity.**

Agronomy is the synthesis of the following disciplines: -

- ✓ Soil science
- ✓ Plant breeding
- ✓ Entomology
- ✓ Crop physiology
- ✓ Plant ecology
- ✓ Plant pathology

Principles Of Agronomy

The important principles of agronomy are **related to cropping system, tillage, crop establishment, nutrient management, water management**, weed management, pest & disease management, harvesting & storage.

Objective

To understand all the components **for their effective management** to get higher productivity.

Scope of agronomy

- ✓ Better **understanding of plant & environment**.
- ✓ **Agricultural practices are modified** to enhance productivity.
- ✓ Appropriate **water and soil management**.
- ✓ **Management of pest & disease of field** crops.
- ✓ To **gain maximum output with minimum** input.

HORTICULTURE

Horticulture is derived from the Latin **word hortus meaning 'garden' and cultura meaning 'cultivation'**. It is a science which deals with the cultivation of plants or garden crops example fruits, vegetables, flower, spices, condiments etc.

Branches of horticulture

Pomology

Pomology is the branch of horticulture that deals with the science of cultivation & management of fruit. **Example – apple, banana, pomegranate etc.**

Olericulture

It is the branch of horticulture that **deals with the cultivation, of vegetable crops**.
Example okra, tomato, potato etc.

Floriculture

This branch of horticulture **deals with the study of cultivation of flowering plants**
example -rose, marigold, carnation, etc.

Viticulture

It is also **called as winegrowing and a branch of horticulture** that deals with the study
of cultivation of wine crops specifically grapes.

Arboriculture

Arboriculture is the cultivation of trees and shrubs especially for ornamental purposes.

Landscape gardening

It is the branch of horticulture **that deals with the architectural arrangements of the garden plant for the purpose of beautification** or ornamentation.

ANIMAL HUSBANDRY

Animal husbandry is the Controlled **cultivation, management, and production of domestic animals, including improvement of the qualities** considered desirable by humans by means of breeding. Animals are raised & breed for utility purpose like food, fibre, sports, power etc.

Branch Of Animal Husbandry

- ✓ **Cattle rearing** -domestication cow for milk, sports and use in field
- ✓ Buffalo domestication
- ✓ **Poultry** – birds' domestication for the purpose of egg & meat
- ✓ **Piggery**- swine/pig domestication
- ✓ **Goat domestication**
- ✓ **Others**- rabbit, horse etc

AGROFORESTRY

Agroforestry is any **sustainable land-use system that maintains or increases total yields by combining food crops (annuals)** with tree crops (perennials) and/or livestock

on the same unit of land, **either alternately or at the same time, using management practices that suit the social and cultural characteristics** of the local people and the economic and ecological conditions of the area.

Example – shifting cultivation, Tungya cultivation etc

Fishery

The production, distribution and consumption of fish and sea food which involve capturing or cultivation of fishes' species in marine or freshwater. **Example rohu, catla, shrimp etc**

Agriculture engineering

The branch of agriculture that deals with the **design of farm machinery, the location and planning of farm structures, farm drainage, soil management and erosion control**, water supply and irrigation, rural electrification, and the processing of farm products